

Notes: Handel & Kolstad (AER, 2015)

Health Insurance for “Humans”: Information Frictions, Plan Choice, and Consumer Welfare

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1 Big picture

- **Question.** How do information frictions, hassle costs, and inertia affect health insurance plan choice, estimates of risk preferences, and welfare analysis?
- **Context.** The firm offers a PPO plan and a high-deductible health plan (HDHP). The HDHP is often financially attractive ex post for many employees because of premiums, employer HSA subsidies, and tax advantages, but relatively few employees choose it.
- **Core puzzle.** Standard insurance models would interpret low HDHP enrollment mainly as evidence of high risk aversion or health risk exposure. The paper argues that this interpretation is incomplete because consumers may misunderstand plan attributes or perceive substantial nonfinancial frictions.
- **Data advantage.** The authors combine administrative data on plan choices and claims with a linked survey measuring knowledge of deductibles, coinsurance, out-of-pocket maxima, HSA features, provider networks, expected hassle costs, total medical expenditure knowledge, and tax-benefit beliefs.
- **Main empirical fact.** Many employees lack basic knowledge about plan financial attributes. For example, only about 27% correctly identify the HDHP deductible, around 19% correctly identify the coinsurance rate, and only about 18% correctly identify the out-of-pocket maximum.
- **Main structural result.** When information frictions, hassle costs, and inertia are added to the model, estimated risk aversion falls sharply relative to a baseline model that interprets all low HDHP take-up through risk preferences alone.
- **Welfare result.** Welfare estimates for a forced switch to the HDHP depend strongly on whether low take-up is interpreted as risk aversion, inertia, information frictions, or hassle costs. Baseline models predict much larger welfare losses than richer models.
- **Policy implication.** Welfare conclusions about high-deductible plans depend on the micro-foundations behind observed choices. Correcting information frictions can improve welfare differently from changing prices or forcing plan enrollment.
- **Takeaway.** Revealed preference from insurance choices can be misleading when consumers are not fully informed or face real and perceived nonfinancial frictions.

2 Incremental contribution of the paper

2.1 What the paper adds relative to standard health insurance choice models

- Standard structural models of health insurance choice typically infer risk preferences from plan choices and claims data, assuming that consumers understand plan attributes and know their own health-risk distribution.
- Handel and Kolstad show that this assumption is empirically questionable in a real employee choice environment.
- Their contribution is not simply to say that consumers are confused; it is to incorporate measured confusion and perceived nonfinancial costs into a structural choice model and show how those measures alter estimates of risk preferences and welfare.

Interpretation: The paper changes the interpretation of insurance choice. Low enrollment in a high-deductible plan is not automatically evidence of high risk aversion or a strong preference for generous insurance. It can reflect low information, misunderstanding of tax benefits, high perceived hassle costs, or default/inertia effects.

2.2 What the paper adds relative to prior evidence on consumer mistakes

- Prior work documents inertia, limited plan knowledge, and mistakes in health insurance and retirement choices.
- This paper links these frictions to administrative claims, realized medical spending, plan menus, and structural welfare calculations.
- The linked survey data allow the authors to observe the exact frictions that would otherwise be absorbed into unobserved heterogeneity or high estimated risk aversion.

Interpretation: The key incremental step is measurement plus modeling: the paper observes consumer knowledge and perceived hassle costs at the individual level, then estimates how those variables change plan valuations.

2.3 What the paper adds for welfare analysis

- Welfare analysis differs depending on whether a choice distortion reflects a preference or a mistake.
- If consumers truly dislike the HDHP because they value risk protection or dislike the plan's nonfinancial features, then forcing them into the HDHP generates large welfare losses.
- If consumers avoid the HDHP because of misinformation or inertia, the revealed preference welfare loss from switching is overstated.
- The paper shows that the welfare cost of a forced HDHP switch is substantially smaller once inertia and information frictions are included.

Economic message: The incremental contribution is a welfare discipline: the same observed plan choices imply very different policy conclusions depending on the micro-foundation behind those choices.

2.4 What the paper adds methodologically

- The paper integrates survey responses directly into a discrete-choice model of health plan selection.
- It uses new employees versus existing employees to help identify inertia, because new employees make active choices without the same default plan.
- It models health risk using detailed claims histories and uses cost simulations to translate plan design into out-of-pocket expenditure distributions.
- It separates baseline risk aversion, inertia, information frictions, time/hassle costs, provider network beliefs, and total medical expenditure beliefs.

Interpretation: The paper provides a template for combining administrative choice data with survey-based measures of beliefs and frictions in structural demand estimation.

3 Data and measurement

3.1 Setting: PPO versus HDHP

- The employer offers a PPO and an HDHP.
- The HDHP has lower premiums and an employer subsidy to a health savings account, but exposes consumers to more out-of-pocket risk and potentially more administrative hassle.
- The plans have comparable provider networks and treatments, so the main financial difference comes from premiums, cost sharing, tax benefits, and HSA features.

- In 2012, only about 17% of employees in the full sample chose the HDHP, even though a much larger share would have been ex post financially better off in the HDHP.

Interpretation: The setting is attractive because the two plans are similar in provider access but differ sharply in financial design and administrative complexity. This helps isolate plan understanding and financial-risk trade-offs.

3.2 Administrative data

- Administrative data include plan choices, demographics, income tier, family size, and claims history.
- The full sample contains 41,361 employees and 115,136 employee-dependent observations.
- The data span plan choices around 2009–2012 and allow the authors to construct realized spending, lagged health risk, and potential out-of-pocket payments in each plan.
- Family spending is highly skewed: Table 1 reports mean family spending around \$10,191, median around \$4,275, and the ninety-ninth percentile around \$87,709 in the full sample.

Interpretation: Administrative data provide the cost side of the choice problem: what each family would pay under each plan in different spending states.

3.3 Survey data

- The survey elicits knowledge of plan financial attributes: deductible, coinsurance, out-of-pocket maximum, employer HSA subsidy, ownership of HSA funds, and the value of tax-free dollars.
- It also measures nonfinancial attributes: provider network beliefs, expected time and hassle costs, feelings about managing health care costs, total medical expenditure knowledge, confidence in that knowledge, and beliefs about tax benefits.
- The survey has a respondent-weighted sample and a reweighted version to make respondents resemble the broader employee population.

Interpretation: The survey allows the authors to distinguish genuine preferences from errors or frictions that are invisible in standard claims and enrollment data.

3.4 Information friction measures

- For each survey question, respondents are classified as correct, incorrect, or not sure.
- For some questions, the authors also separate respondents by plan experience, such as existing HDHP enrollees, new HDHP enrollees, and PPO enrollees.
- The variables are then used as observable shifters in the plan choice model.

Interpretation: The structural model does not merely treat survey responses as descriptive. It uses them to shift the utility of the HDHP relative to the PPO.

3.5 Time and hassle cost measures

- Respondents report expected time spent administering the HDHP.
- They also report whether they understand, accept, or dislike the need to manage health care costs.
- These variables proxy for nonfinancial costs that are not captured by out-of-pocket spending risk.

Interpretation: Time and hassle are especially important because HDHPs can require consumers to monitor balances, understand bills, and manage HSA or cost-sharing mechanics.

4 Conceptual model

4.1 Standard model of insurance choice

The standard insurance model assumes that individual or family k chooses plan j to maximize expected utility:

$$u_{kj} = \int_0^\infty f_{kj}(s \mid \psi_j, \mu_k) u(W_k - P_{kj} - s, \gamma_k) ds. \quad (1)$$

- W_k is wealth.
- P_{kj} is the premium for plan j .
- s is out-of-pocket spending.
- $f_{kj}(s \mid \psi_j, \mu_k)$ is the distribution of spending under plan design ψ_j and family health type μ_k .
- γ_k is risk aversion.

Interpretation: In the standard model, consumers differ in health risk and risk aversion. Low HDHP take-up is therefore interpreted as evidence that consumers strongly value protection from out-of-pocket risk.

4.2 Nonfinancial attributes

The paper extends the model by allowing plan utility to include nonfinancial plan attributes:

$$u_{kj} = \int_0^\infty f_{kj}(s \mid \psi_j, \mu_k) u(W_k - P_{kj} + \pi_j(\psi_j, \mu_k, 1 - t_k) - s, \gamma_k) ds. \quad (2)$$

- $\pi_j(\cdot)$ captures attributes such as provider networks, time and hassle costs, and tax benefits.
- t_k is the marginal tax rate relevant for HSA/FSA benefits.

Interpretation: Plans are differentiated products. Consumers may avoid the HDHP not only because of financial risk but also because of perceived hassle, network concerns, and confusion about tax benefits.

4.3 Information frictions

Consumers may observe plan attributes with error:

$$\hat{\mu}_k = \mu_k + \delta_k^\mu + \epsilon_k, \quad (3)$$

$$\hat{\psi}_j = \psi_j + \delta_j^\psi + \epsilon_j, \quad (4)$$

$$\hat{t}_k = t_k + \delta_k^t + \epsilon_k, \quad (5)$$

$$\hat{\pi}_j = \pi_j + \delta_j^\pi + \epsilon_j. \quad (6)$$

- ϵ is mean-zero error.
- δ captures systematic information frictions or biased beliefs.
- Consumers choose based on perceived, not true, attributes.

Interpretation: With information frictions, observed choices are not clean revealed preferences over true plan attributes. This is why welfare analysis becomes more delicate.

5 Empirical framework

5.1 Baseline choice model

The baseline empirical model uses CARA utility and simulated out-of-pocket spending distributions:

$$u(x; \gamma) = -\frac{1}{\gamma} \exp(-\gamma x), \quad (7)$$

where

$$x_{kj} = W_k - P_{kj} - s + \epsilon_{kj}. \quad (8)$$

Interpretation: The baseline model resembles much of the prior structural health insurance literature. It attributes plan choices mainly to premiums, spending risk, idiosyncratic tastes, and risk aversion.

5.2 Baseline model with inertia

The inertia model adds a utility value from remaining in the incumbent plan:

$$x_{kj} = W_k - P_{kj} - s + \eta(X_k^B) \mathbf{1}\{j_t = j_{t-1}\} + \epsilon_{kj}. \quad (9)$$

- $\eta(X_k^B)$ is an implied monetary switching cost or default effect.
- $\mathbf{1}\{j_t = j_{t-1}\}$ indicates that the consumer stays with the previous plan.
- New employees help identify non-inertial preferences because they choose without the same incumbent default.

Interpretation: Inertia can make consumers stay in dominated or low-value plans even when active preferences would favor switching.

5.3 Full model with frictions

The full model adds survey-based friction measures:

$$x_{kj} = W_k - P_{kj} - s + \eta(X_k^B) \mathbf{1}\{j_t = j_{t-1}\} + Z'_k \beta I_{HDHP} + \epsilon_{kj}. \quad (10)$$

- Z_k contains survey measures of information frictions and hassle costs.
- I_{HDHP} indicates the HDHP plan.
- The coefficients β measure how each friction shifts the value of the HDHP relative to the PPO.

Interpretation: The model allows lack of information or perceived hassle to reduce HDHP demand independently of risk preferences.

5.4 Types model

The authors aggregate frictions into information-type indices. Individuals are sorted into types based on how informed they are.

- The unweighted index counts the number of frictions.
- The weighted index gives more weight to questions more related to choices.
- Type variables shift HDHP value relative to the most informed group.

Interpretation: The types model is more parsimonious than the full model. It asks whether a one-dimensional information index captures the same choice patterns as many separate friction variables.

5.5 Cost model and identification

- Future medical expenditure risk is predicted using detailed prior claims and demographics.
- Plan design maps total medical spending into out-of-pocket spending in each plan.
- Risk preferences are identified from how consumers trade off premiums against out-of-pocket risk.
- Inertia is identified from differences between incumbent employees and new employees.
- Friction coefficients are identified from how survey responses predict HDHP choices conditional on cost risk and other observables.

Interpretation: The empirical design relies on rich variation in health risk, plan choice, default status, and survey responses.

6 Identification and empirical design

6.1 Risk preferences versus information frictions

- A consumer may avoid the HDHP because she is highly risk averse.
- A consumer may also avoid the HDHP because she does not understand the deductible, undervalues HSA tax benefits, overestimates hassle costs, or incorrectly believes the PPO has a larger network.
- The survey variables let the authors separate these channels.

Interpretation: Without survey data, all omitted frictions can be incorrectly loaded into risk aversion or idiosyncratic preference shocks.

6.2 Role of new employees

- Existing employees face a default plan that creates inertia.
- New employees make an active choice without the same incumbent default.
- Comparing similar new and existing employees helps identify choice persistence that is not due to stable preferences.

Interpretation: This variation is important because inertia alone can make consumers remain in plans that do not maximize active-choice utility.

6.3 Why welfare analysis depends on micro-foundations

- If a friction is a real hassle cost, then eliminating it requires reducing that hassle, not simply forcing a plan switch.
- If a friction is misinformation, then providing information can improve choices without forcing enrollment.
- If a choice reflects risk aversion, then forcing a switch may impose large welfare losses.
- If a choice reflects inertia, then active choice policies may have large effects.

Interpretation: The welfare meaning of a coefficient depends on whether it is interpreted as a preference, a mistake, a switching cost, or a hassle cost.

7 Tables and figures

7.1 Table 1 and Figure 1: sample and plan financial trade-off

Read:

- Table 1 describes the administrative and survey samples.
- The full sample contains 41,361 employees and 115,136 employee-dependent observations.
- In 2012, only 17.3% of the full sample chose the HDHP.
- Figure 1 shows that the HDHP has high value for low-spending households because of the subsidy and premium savings, but becomes worse for high-spending households.
- The figure shows that around 60% of employees would be ex post better off in the HDHP under the family tier example.

Economic message: The basic puzzle is visible immediately: the HDHP is financially attractive for many employees, yet relatively few choose it.

Sample demographics	Full sample	Survey recip. (weighted)	Survey resp. (weighted)
Number of employees	41,361	4,500	1,661
N_e : emp. and dep.	115,136	11,690	4,584
2011 PPO%	88.8	89.6	88.7
2012 PPO%	82.7	83.0	81.6
2011 HDHP%	11.2	10.4	11.3
2012 HDHP%	17.3	17.0	18.4
Gender (percent male)	76.4	76.8	75.6
Age (percent)			
18–29	8.6	14.9	11.6
30–39	41.1	43.8	42.7
40–49	38.1	32.7	34.1
50–59	10.9	7.7	10.5
≥ 60	1.3	0.9	1.2
Income (percent)			
Tier 1 (< \$75K)	2.7	2.2	2.2
Tier 2 (\$75K–\$100K)	10.1	13.1	14.0
Tier 3 (\$100K–\$125K)	35.3	38.9	37.9
Tier 4 (\$125K–\$150K)	30.5	29.6	31.3
Tier 5 (\$150K–\$175K)	12.0	10.8	10.0
Tier 6 (\$175K–\$200K)	4.7	3.5	2.9
Tier 7 (\$200K–\$225K)	2.0	1.0	0.9
Tier 8 (\$225K–\$250K)	0.7	0.1	0.0
Tier 9 (> \$250K)	0.8	0.1	0.0
Family size (percent)			
1	23.0	29.0	20.9
2	19.0	19.4	21.9
3+	58.0	51.6	57.2
Family spending			
Mean	\$10,191	\$8,820	\$11,247
Median	\$4,275	\$3,363	\$4,305
25th	\$1,214	\$878	\$1,176
75th	\$10,948	\$9,388	\$11,555
95th	\$35,139	\$32,171	\$41,864
99th	\$87,709	\$80,370	\$87,022

Notes: This table presents summary demographic statistics for the samples we study. The first column represents all employees who were present in our data and have complete records for at least eight months in 2009, 2010, and 2011, and the first month of 2012. The second column represents all employees who received our survey, regardless of whether or not they responded. The third column represents all employees who responded to our survey. Statistics from gender onward represent only 2011, and use the re-weighted statistics for the second and third columns, as described in the text.

Table 1: Sample demographics

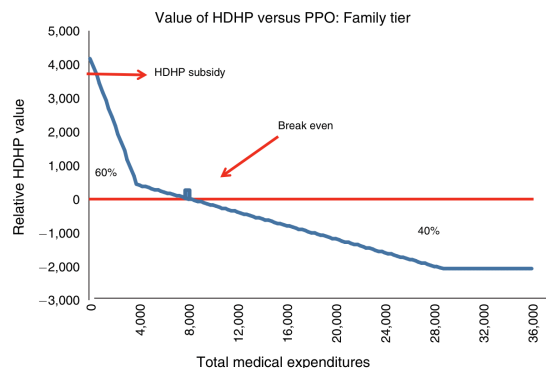


FIGURE 1

Notes: Description of HDHP financial value relative to the PPO in 2012, for the family tier, as a function of total medical expenditures. This chart assumes that employees contribute 50 percent of the maximum possible incremental amount to their HSA, near the median in the population. Sixty percent of all employees break even, given their respective coverage tiers.

Figure 1: Value of HDHP versus PPO

7.2 Tables 2 and 3: measured information and hassle frictions

Read:

- Table 2 shows widespread misunderstanding of HDHP financial features.
- Only 27.08% correctly identify the HDHP deductible, 18.56% the coinsurance rate, and 18.47% the out-of-pocket maximum.

- Knowledge of HSA ownership is much stronger: 75.69% know that unused HSA funds can be kept after the year.
- Table 3 shows major uncertainty about provider networks, time/hassle costs, and own spending.
- Only 34.52% say the provider networks are the same, even though the plans have the same provider network.
- Only 36.66% correctly identify their prior-year expenditure bucket.

Economic message: Consumers lack exactly the information required by standard insurance choice models. This motivates the structural inclusion of frictions.

TABLE 2—RESPONSES TO PLAN FINANCIAL CHARACTERISTICS SURVEY QUESTIONS (Percent)

Question	Correct	Incorrect	Not sure
1. What is the deductible under the HDHP?	27.08	22.40	50.53
HDHP: existing	52.68	11.23	36.10
HDHP: new	50.79	13.49	35.73
PPO	21.53	24.66	53.82
2. What is the co-insurance rate under the HDHP?	18.56	25.64	55.80
HDHP: existing	33.85	21.24	44.91
HDHP: new	29.07	21.37	49.56
PPO	15.66	26.61	57.73
3. What is the out-of-pocket maximum under the HDHP?	18.47	21.98	59.55
HDHP: existing	28.32	22.11	49.57
HDHP: new	31.87	18.91	49.21
PPO	15.85	22.31	61.84
4. How much is the employer HDHP subsidy?	31.42	19.94	48.64
HDHP: existing	73.40	11.05	15.54
HDHP: new	68.65	11.21	20.14
PPO	22.50	21.92	55.58
5. Do you get to keep HSA funds after the end of the year?	75.69	9.23	15.08
HDHP: existing	96.73	1.38	1.90
HDHP: new	94.22	1.75	4.03
PPO	71.23	10.96	17.81
6. How much is \$1,000 worth in pretax dollars?	14.50	44.86	40.64
HDHP: existing	16.93	31.78	51.30
HDHP: new	15.76	42.73	41.51
PPO	14.09	46.58	39.33

Note: Exact wording of questions and answers in online Appendix A.

Table 2: Financial knowledge questions

Question	Same	HDHP bigger	PPO bigger	Not sure			
7. How do the provider networks of the two plans compare?	34.52	6.04	12.46	46.98			
HDHP: existing	41.28	6.74	2.76	49.22			
HDHP: new	49.39	3.33	4.20	43.08			
PPO	32.09	6.26	14.48	47.16			
	None	< 1 hour	1–5 hours	6–10 hours	11–20 hours	> 20 hours	Not sure
8. How much time do you expect to spend in the HDHP?	1.76	5.99	21.73	17.40	12.88	24.92	15.34
HDHP: existing	5.18	19.17	46.11	17.62	5.53	6.39	—
HDHP: new	3.50	14.71	40.81	22.24	11.21	7.53	—
PPO	1.17	3.52	16.83	16.83	13.89	28.96	18.79
In the PPO?							
PPO	15.85	29.75	29.16	11.35	2.94	4.11	6.85
	Understand, not concerned	Accept, but concerned	Don't like, no matter what				
9. How do you feel about spending time managing your health plan?	14.82	42.52	42.65				
HDHP: existing	39.03	32.64	28.32				
HDHP: new	26.62	39.05	34.33				
PPO	10.76	44.04	45.21				
	Correct	Overestimate	Underestimate	Not sure			
10. How much was spent on you and your dependents in 2011?	36.66	29.81	23.31	10.22			
HDHP: existing	41.97	35.75	16.41	5.87			
HDHP: new	37.13	27.85	23.47	11.56			
PPO	36.01	29.35	24.07	10.57			
	Very confident	Somewhat confident	Not confident				
11. How confident are you in this estimate?	35.85	43.90	20.25				
HDHP: existing	38.34	49.22	12.44				
HDHP: new	30.11	46.13	23.77				
PPO	36.20	43.05	20.74				
	Yes	No	Not sure				
12. Do you think you will benefit/would have benefited from the HDHP in 2012?	16.49	58.35	25.16				
HDHP: existing	56.65	23.83	19.52				
HDHP: new	30.47	42.91	26.62				
PPO	10.37	63.99	25.64				

Note: Exact wording of questions and answers in online Appendix A.

Table 3: Nonfinancial and hassle frictions

7.3 Figures 2 and 3: choices by network and hassle perceptions

Read:

- Figure 2 shows that people who believe the PPO provider network is larger are much less likely to choose the HDHP.
- Figure 3 shows that expected time and hassle costs are strongly related to HDHP take-up.
- The HDHP share is highest among people expecting no time/hassle costs and lowest among those expecting more than 20 hours.

Economic message: Beliefs about nonfinancial plan features predict actual choices. This supports the model's use of survey responses as choice shifters.

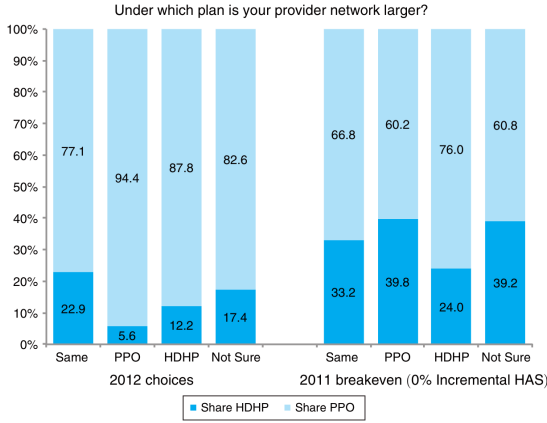


FIGURE 2. ACTUAL VERSUS PREDICTED PLAN CHOICES BY KNOWLEDGE OF PLAN PROVIDER NETWORKS

Figure 2: Choice by provider-network beliefs

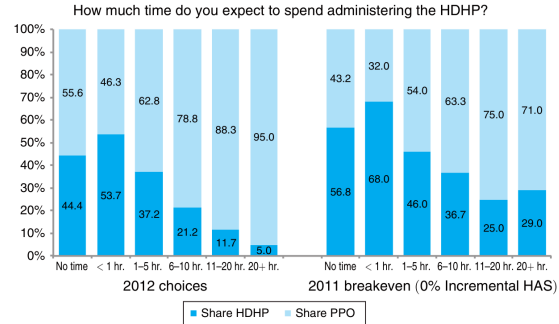


FIGURE 3. ACTUAL VERSUS PREDICTED CHOICES AS A FUNCTION OF TIME AND HASSLE COST PERCEPTIONS

Figure 3: Choice by expected time and hassle costs

7.4 Figure 4 and Tables 4–5: information types and structural estimates

Read:

- Figure 4 shows the distribution of the information-type index q .
- Most respondents are neither fully informed nor fully uninformed; the distribution has substantial mass in middle information categories.
- Table 4 shows that estimated risk aversion falls sharply as inertia and frictions are added.
- The gamble interpretation of average risk aversion rises from \$366.74 in the baseline to above \$900 in richer models, implying much lower estimated risk aversion.
- Table 5 shows that type-index specifications deliver similar economic messages: less-informed types have substantially lower HDHP valuations.

Economic message: Incorporating information and hassle measures changes the estimated preference parameters. What looked like risk aversion in a simple model is partly misinformation, inertia, and hassle.

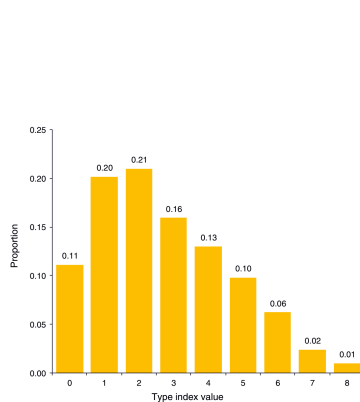


FIGURE 4. HISTOGRAM OF INFORMATION TYPE INDEX q FOR THE SAMPLE OF SURVEY RESPONDENTS

Figure 4: Information-type index

Primary model estimates	Base (1)	Base + inertia (2)	Full model (3)	Structural information (4)
Average μ_i	$1.6 \cdot 10^{-37}$	$2.3 \cdot 10^{-4}$	$8.6 \cdot 10^{-35}$	$4.9 \cdot 10^{-32}$
SD μ_i	$3.1 \cdot 10^{-41}$	$3.6 \cdot 10^{-5}$	$1.4 \cdot 10^{-39}$	$6.16 \cdot 10^{-35}$
Gamble interp. of average μ_i	366.74 ^a	812.61	920.47 ^a	952.89 ^a
σ_i	$1.8 \cdot 10^{-39}$	$1.6 \cdot 10^{-4}$	$2.2 \cdot 10^{-37}$	$5.8 \cdot 10^{-34}$
Total SD γ	$2.0 \cdot 10^{-37}$	$1.7 \cdot 10^{-4}$	$1.4 \cdot 10^{-39}$	$8.5 \cdot 10^{-35}$
$\sigma_{i, HDHP}$	149.23	5.01	0.11	0.13
Benefits knowledge:				
Any incorrect	—	—	98.04	—
Any “not sure”	—	—	467.48	—
Time cost hrs, X pref:				
Time cost hrs:	—	—	-9.72	-12.31
X accept, concerned	—	—	-118.15**	-195.11**
X dislike	—	—	-128.98**	-220.38**
Provider networks:				
HDHP network bigger	—	—	-594.38	-1,424.55
PPO network bigger	—	—	-2,362.85**	-4,513.65**
Not sure	—	—	-301.81	-507.00
TME guess:				
Overestimate	—	—	62.98	1,178.21**
Underestimate	—	—	-208.30	-1,175.57
Not sure	—	—	-688.91	-1,552.80**
Average survey effect	—	—	-1,787.40**	-3,022.12**
σ survey effect	—	—	1,303.64	2,617.07
Likelihood ratio	—	—	379.54	—
Test stat versus (2)	—	—	—	—

Notes: This table presents the estimates for our primary models of interest described in the text, plus a more structural version of the full model, described in more depth in online Appendix D. Column 1 presents the baseline model results (health risk, risk preferences) while column 2 presents the results for the baseline model with inertia. Column 3 presents the results of our full model, which includes all information frictions and hassle costs, as described in Section III. Column 4 presents the results of the full model that treats responses to information-related questions structurally. Standard errors for all parameters presented in online Appendix E.

^aPoint estimate outside of 95 percent CI for same parameter in model (2).

Aggregated information types and hassle costs	Base + inertia (2)	Full model (3)	Types unweighted (5)	Types weighted (6)
Average μ_i	$2.3 \cdot 10^{-4}$	$8.6 \cdot 10^{-35}$	$7.4 \cdot 10^{-35}$	$8.9 \cdot 10^{-35}$
SD μ_i	$3.6 \cdot 10^{-5}$	$1.4 \cdot 10^{-39}$	$1.6 \cdot 10^{-37}$	$1.1 \cdot 10^{-35}$
Gamble interpretation	812.61	920.47 ^a	930.64 ^a	917.83 ^a
σ_i	$1.6 \cdot 10^{-4}$	$2.2 \cdot 10^{-37}$	$4.9 \cdot 10^{-36}$	$2.1 \cdot 10^{-35}$
Total SD γ	$1.7 \cdot 10^{-4}$	$1.4 \cdot 10^{-39}$	$1.7 \cdot 10^{-37}$	$1.1 \cdot 10^{-35}$
$\sigma_{i, HDHP}$	5.01	17.70	0.07	1,828.25
Unweighted information index ^a	—	—	—	—
Lowest quartile	—	—	-6,803.50**	—
Second quartile	—	—	-3,547.10**	—
Third quartile	—	—	-1,158.95**	—
Weighted information index ^a	—	—	—	—
Lowest quartile	—	—	—	-3,655.12**
Second quartile	—	—	—	-1,928.53**
Third quartile	—	—	—	-49.46
Time cost hrs, X pref:				
Time cost hrs:	—	—	-9.72	-33.55
X accept, concerned	—	—	-118.15**	-119.99**
X dislike	—	—	-128.98**	-140.73**
Provider networks:				
HDHP network bigger	—	—	-594.38	-1,424.55
PPO network bigger	—	—	-2,362.85**	-4,513.65**
Not sure	—	—	-301.81	-507.00
TME guess:				
Overestimate	—	—	62.98	1,178.21**
Underestimate	—	—	-208.30	-1,175.57
Not sure	—	—	-688.91	-1,552.80**
Average survey effect	—	—	-1,787.40**	-3,022.12**
σ survey effect	—	—	1,303.64	2,617.07
Likelihood ratio	—	—	379.54	—

Notes: This table presents the results from the “type” models that aggregate our measures of information frictions into one-dimensional indices that describe the level of information an individual has. Section III in the text describes our two type measures “unweighted” and “weighted” in more detail. The last two columns in the table present the type models, while the first two columns, for comparison purposes, restate the results from Table 4—of (i) the primary model with inertia but no frictions and (ii) the full model. The overall implications for how risk preference estimates change are similar with the type models and the full models, suggesting the type measures are a good representation of underlying heterogeneity in information frictions. Moreover, as expected, the more informed types make “better” choices on average and are more likely to value the high-deductible plan appropriately. Standard errors for all parameters presented in online Appendix E.

^aQuintile category is the fourth quartile, i.e., the most informed consumers.

^bPoint estimate outside of 95 percent CI for same parameter in model (2).

**95 percent CI for parameter does not include 0.

Table 5: Types models

7.5 Table 6 and Figures 5–6: welfare and moral hazard

Read:

- Table 6 reports welfare effects of a forced switch to the HDHP.
- The baseline model without inertia predicts a mean welfare loss of \$1,237.61.
- The baseline model with inertia predicts a smaller mean loss of \$874.46.
- The full model predicts a mean loss of about \$788.94.
- The risk-neutral benchmark predicts a mean loss of \$726.09.
- Figure 5 shows distributional welfare effects across percentiles.
- Figure 6 shows the minimum moral-hazard elasticity needed to justify the forced switch based on reduced medical spending.

Economic message: Welfare conclusions differ sharply across models. Adding realistic frictions narrows the gap between structural welfare estimates and the risk-neutral benchmark.

TABLE 6						
Forced HDHP enrollment welfare analysis						
Model	Mean	SD	25th	Median	75th	95th
Baseline model, no inertia	-1,237.61	851.31	-1,833.91	-1,422.06	-694.33	288.57
Baseline model	-874.46	975.27	-1,691.54	-953.03	-238.38	793.72
Full model, no inertia	-788.33	1,021.62	-1,651.19	-866.79	-109.89	971.53
Full model	-788.94	1,021.47	-1,653.4	-867.61	-114.01	968.42
Structural frictions	-741.75	1,060.47	-1,596.03	-804.55	-2.51	1,115.76
Risk neutral	-726.09	1,056.82	-1,622.6	-791.78	-24.150	1,120.93
Moral hazard necessary to justify switch						
	Elasticity lower bound		Elasticity upper bound			
Baseline model, no inertia	0.280		0.407			
Baseline model	0.197		0.286			
Full model	0.178		0.258			
Risk neutral	0.164		0.237			

Notes: The top half of this table presents the welfare impact of a menu redesign that removes the PPO option and forces all consumers into the HDHP. The welfare results are presented for each of five different models to illustrate the impact of incorporating inertia, information frictions, and hassle costs on top of a basic model with health risk and risk preferences. The bottom half of the table illustrates, for each potential underlying model, the minimum consumer price elasticity of demand for medical expenditures that can generate enough cost savings to justify the forced switch to the HDHP.

Table 6: Welfare effects of forced HDHP switch

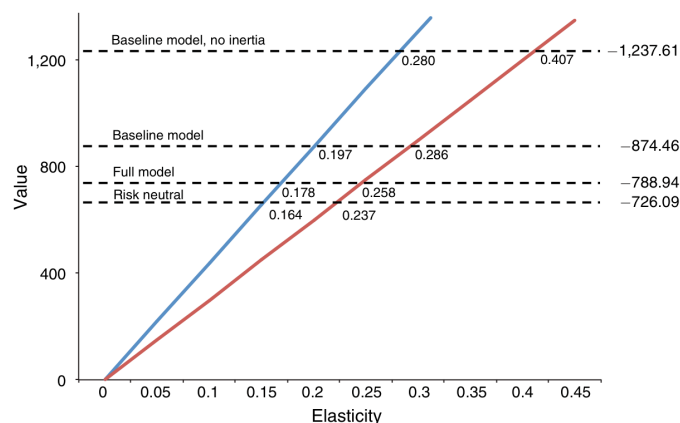


FIGURE 6

Notes: This figure describes the minimum price elasticity necessary to justify the forced switch to the HDHP for each of the choice models studied. The horizontal lines represent the mean consumer surplus change due to increased risk exposure for the different models studied. The upper diagonal line shows the relationship between total cost savings (y-axis) from reduced expenditures in the HDHP for a given price elasticity of utilization ξ (x-axis) when forgone medical expenditures are assumed to be "purely wasteful." The second diagonal line shows the relationship between the mean of the cost savings from the HDHP and the mean of the cost savings from the HDHP when the cost savings are assumed to be "purely wasteful."

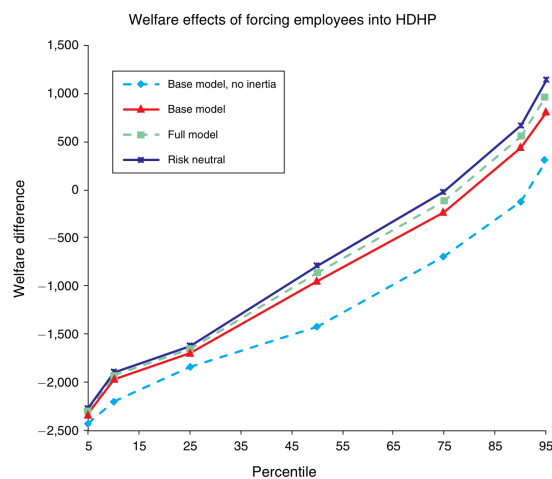


FIGURE 5

Notes: This figure plots quantiles of the welfare impact of the forced HDHP switch for each of the four models presented in Table 6. The results for both the types model and the full model

Figure 5: Distributional welfare effects

8 Short summary

- The paper studies health insurance plan choice using linked administrative and survey data.
- The standard model treats choices as revealed preferences over financial risk.
- Many employees lack information about deductibles, coinsurance, out-of-pocket maxima, tax benefits, provider networks, and total spending.
- Perceived time and hassle costs are strongly related to HDHP choice.
- Adding inertia and survey-measured frictions substantially reduces estimated risk aversion.
- The welfare cost of forcing employees into the HDHP is smaller in richer models than in the baseline model.
- The paper demonstrates that welfare analysis requires knowing why consumers choose as they do, not merely what they choose.

9 Conclusion

9.1 Core conclusion

Handel and Kolstad show that health insurance choices cannot be interpreted reliably using a simple model of fully informed risk-averse consumers. Information frictions, perceived hassle costs, and inertia are important determinants of plan choice and materially change risk-preference estimates and welfare calculations.

9.2 Economic intuition

Consumers do not simply choose between premium savings and financial risk. They also choose under uncertainty about plan rules, provider networks, tax benefits, health expenditures, and the administrative burden of using a high-deductible plan. If the researcher ignores these frictions, the model treats low HDHP enrollment as high risk aversion. Once frictions are measured directly, much of this apparent risk aversion is reinterpreted as misinformation, inertia, or hassle.

9.3 Incremental contribution takeaway

The paper's incremental contribution is to bring consumer-information measurement into structural insurance demand estimation. It shows that the welfare implications of plan redesign depend on whether observed choices are preferences or mistakes. This contribution is central for evaluating high-deductible plans, active choice policies, consumer information interventions, and health insurance market design.